

Phosphate plant -- Control

Section report: Control

Site	Atlas Phosphate Complex (:atlas_phosphate)
Reporting period	2026-01-01T00:00:00 to 2026-12-31T23:00:00 (365 days, 8568 operating hours)
Product	Dap from 29.0 % P2O5 ore, 1713.0 kt of product
Framework scope	ISO 50001:2018 energy management; ISO 14001:2015 environmental management; ISO 46001:2019 water efficiency; ISO 9001:2015 quality management; GHG Protocol corporate standard; EU Industrial Emissions Directive 2010/75; US EPA 40 CFR 63 subpart AA; IFA/EFMA BAT for phosphate fertilisers; Phosphogypsum stack management code
Provenance	Design revision A seed 20260101 8760 hourly intervals of 69 signals generated 2026-09-23T15:43:11.110 basis :screening_plant_model

Predictive control of the reaction train

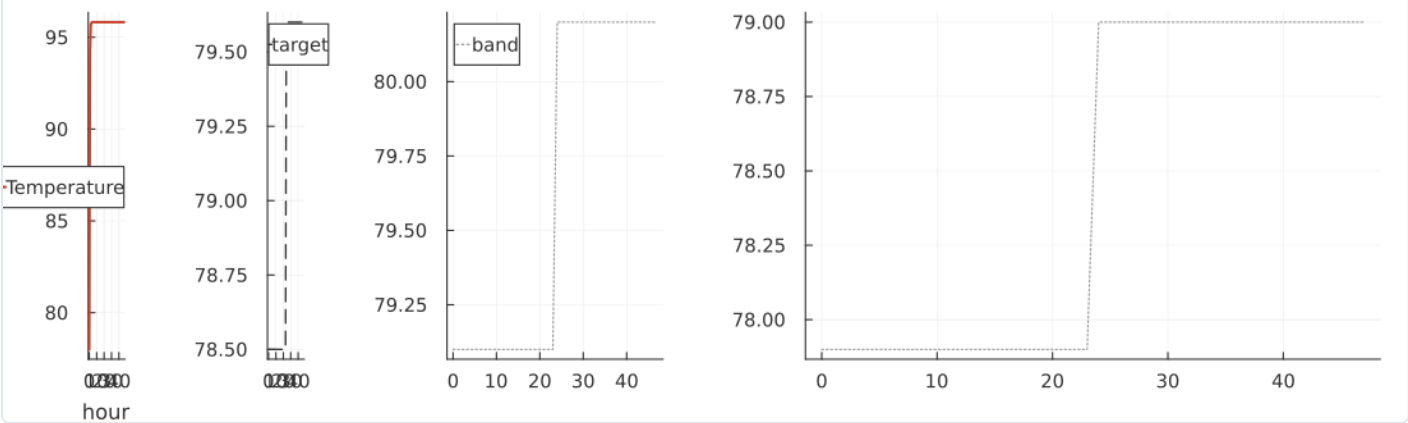
The controller is the one the plant would run: a linear predictive controller over the discrete model 11 x 11 obtained by linearising the ODE twin at its operating point (12 prediction steps, 4 control steps, sample time 1 h), with a Kalman filter estimating the states from the measurements. The plant of the test is the *nonlinear* twin, so the controller never sees the equations it is controlling.

CLOSED LOOP OF THE TWO CONTROLLERS

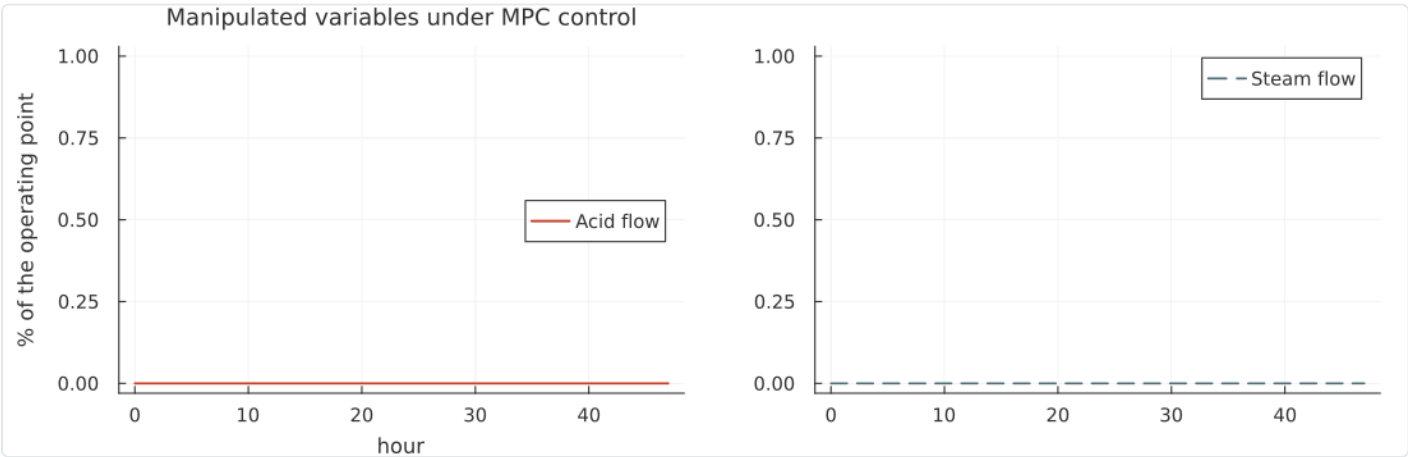
Quantity	Value
Scenario	:setpoint over 48 hours
Model	11 states, 6 inputs, 12 outputs, Ts = 1 h
Controllability	conditioning 3.4124288612524188e16
Poles	largest magnitude 1
MPC error	14,863 bands and 284 violations
PI error	2,867.68 bands and 233 violations

Controlled variable	Band	MPC error	PI error	Improvement %	MPC worst	PI worst	MPC hours out	PI hours out
:temperature	0.6	1,310.92	41.86	-3,031.32	28.87	0.9654	47	0
:free_so4	0.2	8,282.93	604.12	-1,271.06	195.69	17.27	47	42
:acid_strength_weak	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
:acid_strength	0.15	4,041	942.67	-328.68	89.91	23.24	48	48
:product_rate	6	1,102.69	1,115.28	1.13	23.35	23.35	48	48
:filter_rate	0.35	70.65	60.84	-16.12	1.48	1.27	48	48
:mill_power	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
:p80	6	54.82	102.9	46.73	1.17	2.19	46	47
:bed_moisture	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
:granulator_temperature	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
:mill_inventory	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
:fouling_pct	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a

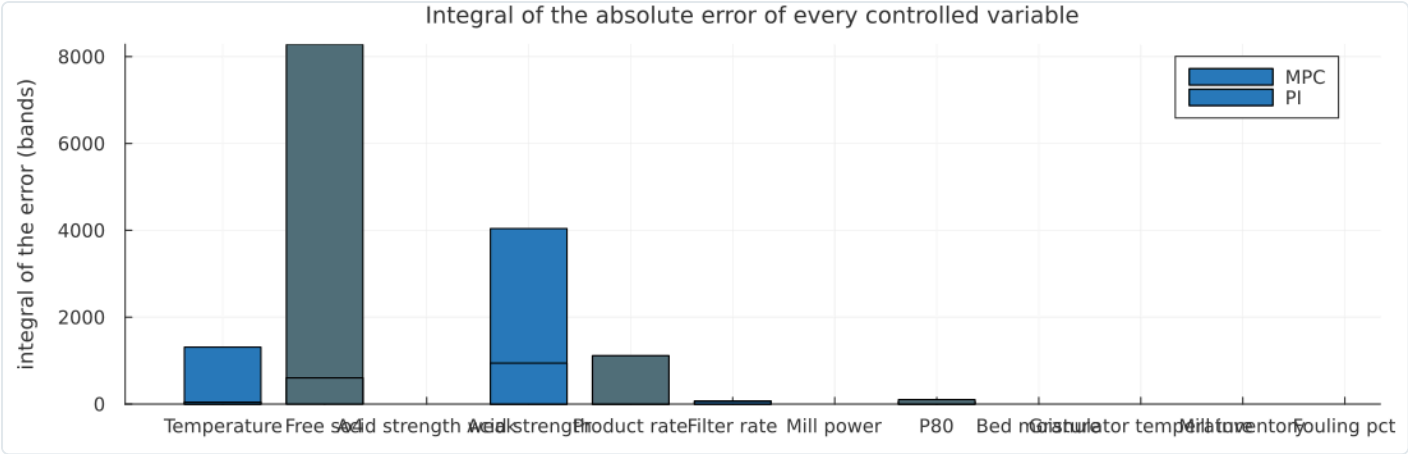
Temperature under MPC control (:setpoint)



Closed loop of the predictive controller



Manipulated variables the predictive controller used



Integral of the absolute error of the two controllers, output by output

WORK THE TWO CONTROLLERS ASKED OF THE PLANT

Manipulated variable	MPC movement %	PI movement %	MPC reversals	PI reversals	MPC range %	PI range %	
:acid_flow		0	14.94	0	0	0	14.94
:ammonia_flow		0	0	0	0	0	0
:cooling_water		0	40	0	0	0	40
:mill_feed		0	2.62	0	0	0	2.62
:steam_flow		0	17.75	0	0	0	17.75
:wash_water		0	0	0	0	0	0